ARMORER'S BENCH TECHNICAL MANUAL

Kalashnikov AK-47 / AKM / AK-74

Izhmash / Tula / Cugir / Arsenal / Century • Stamped & Milled Automatic Rifles (7.62x39mm & 5.45x39mm Soviet)

Manufacturer:	Izhmash / Tula / Cugir / Arsenal / Century	**Category:**	Rifles & Carbines
Action Type:	Long-Stroke Gas Piston Semi-Automatic	**Era / Service:**	Global Combat Legend (1947–Present)
Calibers:	7.62x39mm Soviet, 5.45x39mm, 5.56x45mm NATO	**Covered Models:**	AK-47 Type 3 (Milled), AKM (Stamped 1mm), AK-74 (5.45mm), RPK Light Machine Gun, AK-103, WASR-10

KEY ARMORER TASKS & BENCH OPERATIONS

Core Procedures: Front Trunnion Rivet Forming (3-ton squeeze) • Barrel Pin Pressing & Headspace Setting • Bolt Locking Lug Lapping • Trigger Group Shepherds Hook Servicing • Gas Tube Retainer Lever Fitting

CRITICAL ARMORER TOLERANCES & SPECIFICATIONS

SPECIFICATION / CLEARANCE	ARMORER TOLERANCE & BENCH SPECIFICATION
HEADSPACE RULE	Measured with completely stripped bolt (extractor removed). Bolt MUST close on 7.62x39 GO (1.3032"); MUST NOT close on NO-GO (1.3090"); closing on FIELD (1.3140") is UNSAFE
BARREL PIN PRESS FIT	Transverse 7mm barrel pin requires 2.5 to 3.5 tons hydraulic press force to insert through trunnion
RIVET CRUSH SPEC	Front trunnion uses 4mm swell-neck rivets crushed into chamfered receiver holes creating shear-proof bond
GAS SYSTEM CLEARANCE	Long-stroke piston pinned to carrier; piston must have intentional wobble (approx 0.010"–0.020" play) to self-align in gas tube
TRIGGER RETAINER	Shepherd's hook wire or aftermarket red star retaining plate captures hammer and trigger axis pins

IMMEDIATE ACTION & MALFUNCTION DIAGNOSTIC MATRIX

MALFUNCTION CONDITION	PROBABLE CAUSES	ARMORER BENCH REMEDY
Stoppage 1: Trigger Slap (Severe Vibration into Shooter Trigger Finger) **Firing a round causes violent painful sting/kick against the shooter's trigger finger**	• Disconnector tail contacting the hammer spur during recoil travel. • Improperly reprofiled commercial trigger group in converted imported sporter AK.	**Gunsmith Action: Disassemble trigger group. Remove hammer and disconnector. Grind 0.050" off bottom rear tail of disconnector where it contacts the rear trigger foot. Polish contact surfaces to ensure hammer clears disconnector without bottoming out.**
Stoppage 2: Bolt Hangup on Hammer Face (Carrier Binds Mid-Stroke) **Manually racking bolt carrier feels rough; carrier gets stuck held open on top hump of hammer**	• Commercial replacement hammer has sharp angular profile instead of gentle radius. • Rough machining tool marks on bottom sliding rails of bolt carrier.	**Gunsmith Action: Stone and blend the top radius of the hammer face to match original Soviet military geometry. Polish hammer face with 600-grit paper. Apply light dab of grease to carrier underside track.**

CURATED BENCH VIDEO REFERENCES

PROCEDURE TITLE	INSTRUCTOR / CREATOR	TECHNICAL PROCEDURE SUMMARY
AK-47 / AKM Complete Armorer Inspection, Rivets & Headspace	Jim Fuller / Rifle Dynamics	Master AK builder Jim Fuller explains front trunnion riveting, pressing barrel pins, setting headspace with stripped bolts, and tuning trigger groups.